

lenses lose elasticity and flexibility. As the lens in the eye becomes stiff, muscles near it begin to contract and cannot change the shape of the lens for bringing nearby objects into focus.

Devesh Mistry's invention needs liquid crystal (which is required for LCD used in TV monitors, smartphones and tablets), that can make the eye lenses adjust and focus depending on the eye muscle movement automatically. His research is based on developing synthetic replacements for a diseased lens in our eye, that is, intra-ocular lens implants that can bring back normal sight for the elderly.

Wearable sensors could translate sign language into English

Wearable sensors could one day interpret the gestures in sign language and translate these into English language, providing a solution to communication problems between deaf people and those who do not understand sign language.



Prototype of a wearable device that can sense movement and muscle activity in a person's arm (Image courtesy: www.mytechocean.com)

Engineers at Texas A&M University, USA, are developing a wearable device that can sense movement and muscle activity in a person's arms. It works by figuring out the gestures a person is making by using two distinct sensors: one that responds to the motion of the wrist and the other to the muscular movements in the arm. A program then wirelessly receives this information and converts the data into English.

The researchers built a prototype system that could recognise words that people use most commonly in their daily conversations. They say that once the team starts expanding the program, they will include more words that are less frequently used, in order to build up a more substantial vocabulary.

Genetically-engineered virus doubles efficiency of solar cells

Researchers at MIT have developed a genetically-engineered virus that could help solar cells generate energy with nearly double efficiency.

Plants undergo photosynthesis where these absorb

sunlight and convert it into solar energy with almost 100 per cent efficiency. In the case of solar panels, even the best can only convert about 44 per cent of the absorbed light into usable energy.

One way plants achieve this efficiency is by using a concept called quantum weirdness that allows a particle to exist in more than one place at a single instant of time.

Researchers took advantage of this quantum weirdness by changing viruses' DNA, which improved the speed of excitons by more than 200 per cent and travelled at double the speed of those in existing solar cells, thus resulting in solar panels that could transmit energy with unprecedented efficiency.

Scientists develop world's first 3D blood vessel bio-printer

A Chinese company has developed the world's first 3D blood vessel bio-printer, which makes it possible to produce personalised functional organs.

According to biotechnological company Sichuan Revotek Co. Ltd, based in Sichuan Province, the breakthrough has been achieved through its self-developed stem cell bio-ink technology, 3D bio-printer and cloud computing platform.

James Kang, who led the programme, said that blood vessels that transport nutrients to organs are indispensable elements when creating any organs. His team created a novel type of bio-ink—Biosynsphere—whose primary goal is the personalised stem cell bio-printing to pave the way for organ regeneration.

3D-printed swimsuit scrubs water pollution

When most people swim in the ocean, they do not think about how they can mop up pollution as they splash around, but a team of researchers have created a 3D-printed bathing suit that could do just that.

Engineers from University of California, USA, along with designers from Eray Carbajo, an architecture and design firm based in New York City, have designed a bikini swimsuit that can absorb contaminants from water while a person swims. The swimsuit keeps the materials locked up in its fabric.

The suit has been moulded from a 3D-printed, net-like structure made of synthetic rubber and a reusable and recyclable padding made from a material called Sponge, which was also developed by the team four years ago. They were trying to create a useful technology to help clean up oil or chemical spills, or desalinate water.

They created Sponge from heated sucrose (compound found in sugarcane) that was modified on the molecular level. It is highly-porous but it is also hydrophobic, which means it does not absorb water. If you put Sponge in an oil and water mixture, it will absorb the oil but not water.

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